

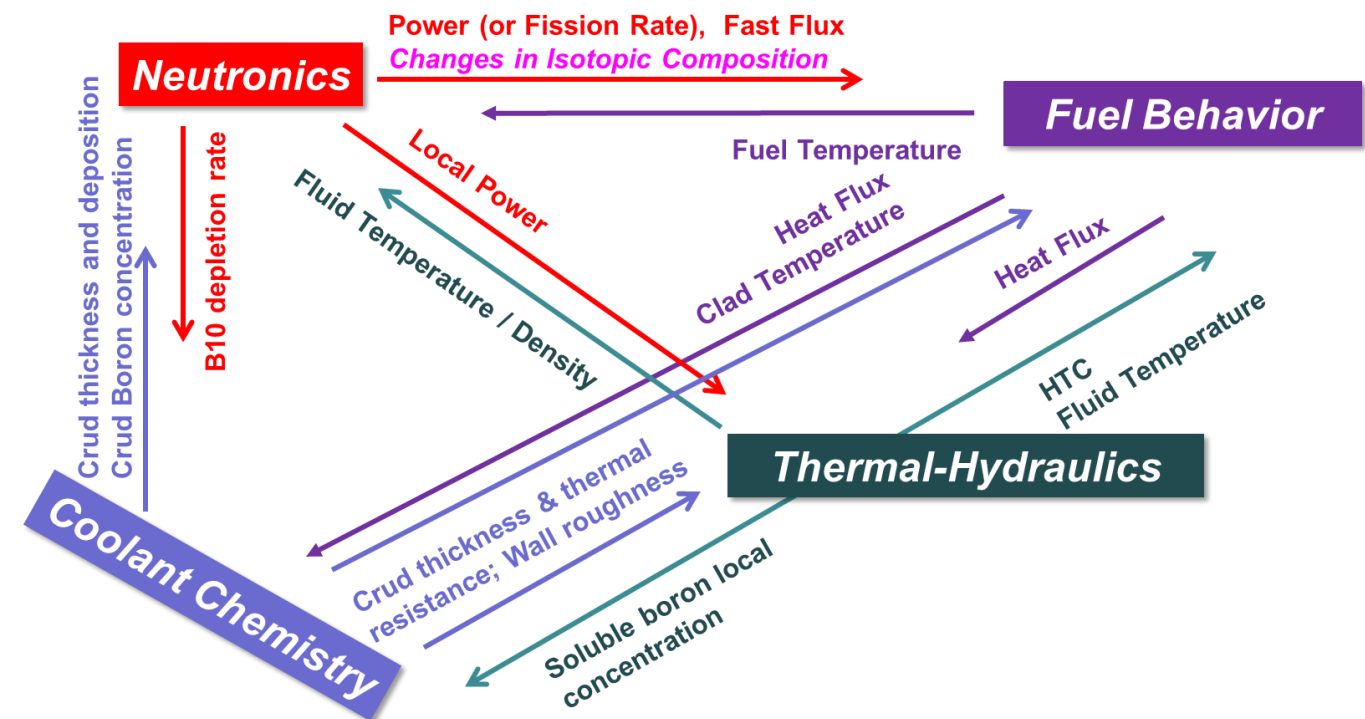
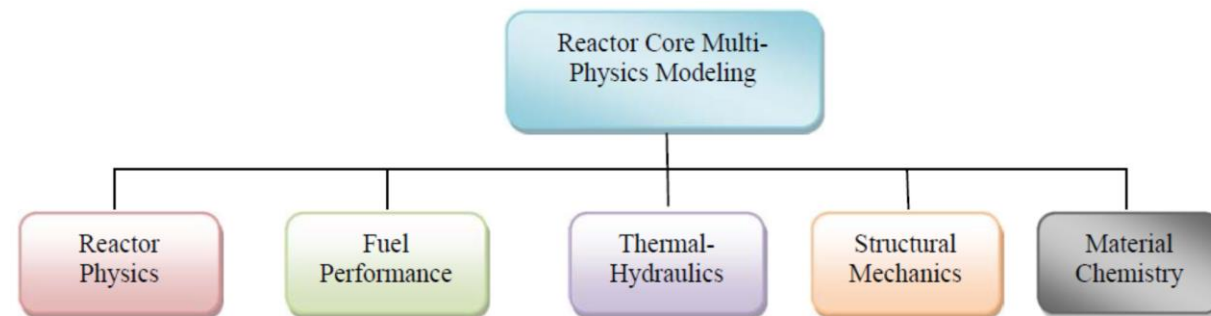
NE: Multi-Physics of Nuclear Reactors

UNIT 3: Low Fidelity Multi-Physics Modeling

Lecture 13: COUPLED THERMAL-HYDRAULICS (INCLUDING FUEL) AND NEUTRONICS SIMULATIONS

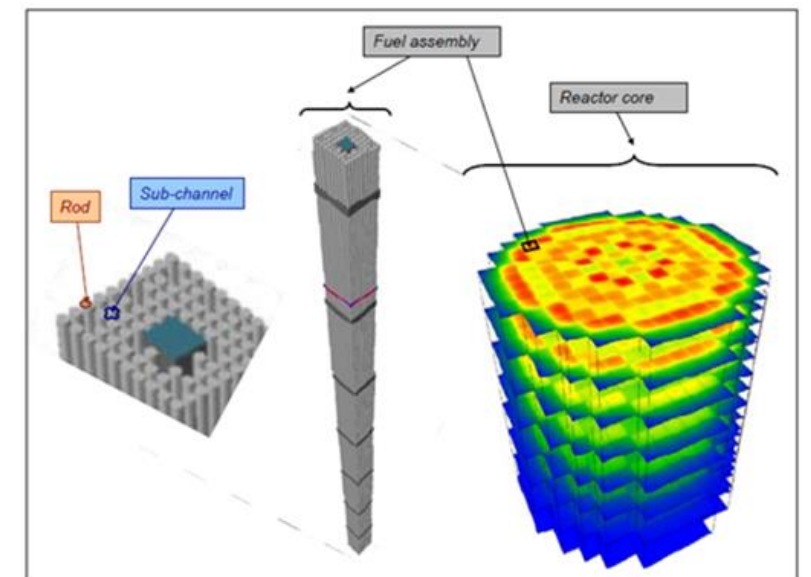
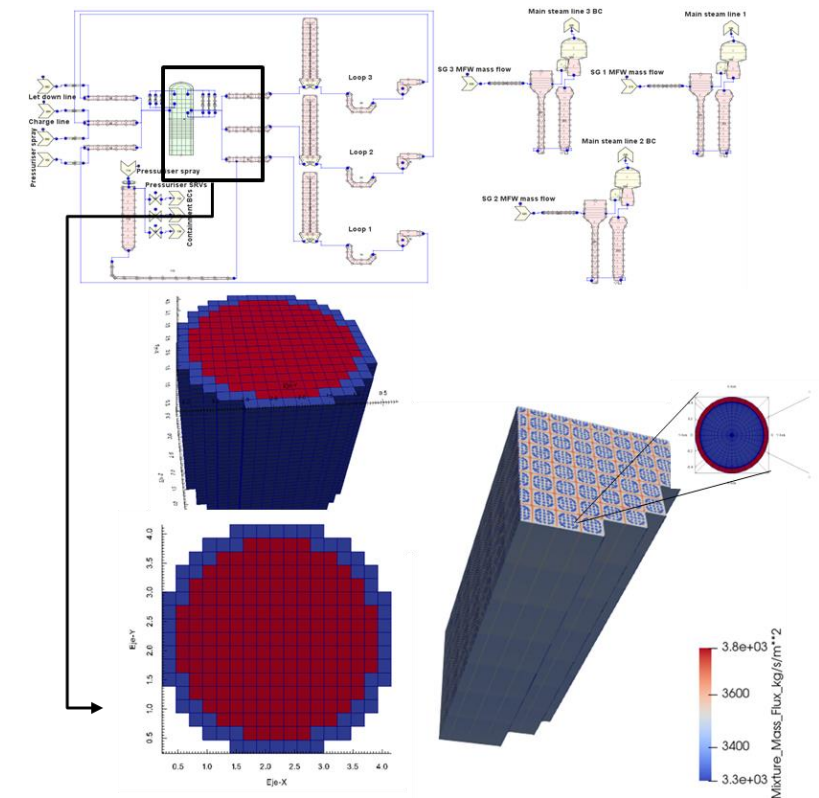
INTRODUCTION AND BACKGROUND

- Single physics modeling and simulation focuses on single physics phenomenon such as nuclear fission and considers the interactions with other physics phenomena via boundary conditions.
- Multi-physics modeling and simulation describe nonlinear multi-physics phenomena by direct coupling of feedback effects of different physics.



CLASSIFICATION OF MULTI-PHYSICS MODELING

- The reactor core multi-physics coupling can be carried out at different scales (levels) in spatial domain:
 - assembly (sub-assembly)/channel level;
 - pin/sub-channel level;
 - sub-pin (pin-resolved) heterogeneity level;
 - or combinations of the above-described levels.
- Recent OECD/NEA classification has identified three major types of M&S tools:
 - Traditional low-fidelity, low-resolution tools used in nuclear industry and regulation for routine calculations;
 - Novel high-fidelity, high-resolution codes used mostly for reference solutions of challenge problems;
 - Tools using high-to low (hi2lo) model fidelity information to combine accuracy with computation efficiency.



PRE-TRADITIONAL MULTI-PHYSICS MODELING

- Timeline: up to early 1990s
- Limitations: computational resources
- Simplifications: decoupled multi-physics simulations
 - *Only the physics of interest treated in detail*
 - *Remaining physics represented by*
 - ***simplified models:***
 - core physics simulators: 1-D thermal-hydraulics models and 1-D fuel conduction
 - thermal-hydraulics codes: point kinetics and 1-D fuel conduction
 - fuel performance codes: 1-D neutronics and depletion equation and 1-D thermal-hydraulics models
 - ***boundary conditions:*** tabulated total power and local power distributions in thermal-hydraulics codes
 - ***parameterized or fixed values:*** fuel temperatures tables (as function of power and burnup) for neutronics codes

TRADITIONAL MULTI-PHYSICS MODELING

- Timeline: currently used in nuclear industry and regulatory practice
- The first traditional multi-physics coupling involved
 - coupling 3-D reactor physics (kinetics) with core thermal-hydraulics; and
 - coupling 3-D core models with system thermal-hydraulics.
- Later, system models were coupled with containment models.
- The traditional multi-physics tools, although updated for today's technology, still suffer from the limitations of their original design intent to approximate or fit existing data by means of empirical models.

COUPLING METHODOLOGIES

Objective - to provide accurate solutions in a reasonable amount of CPU time.

Capability - simulations of detailed operational transients and accident scenarios.

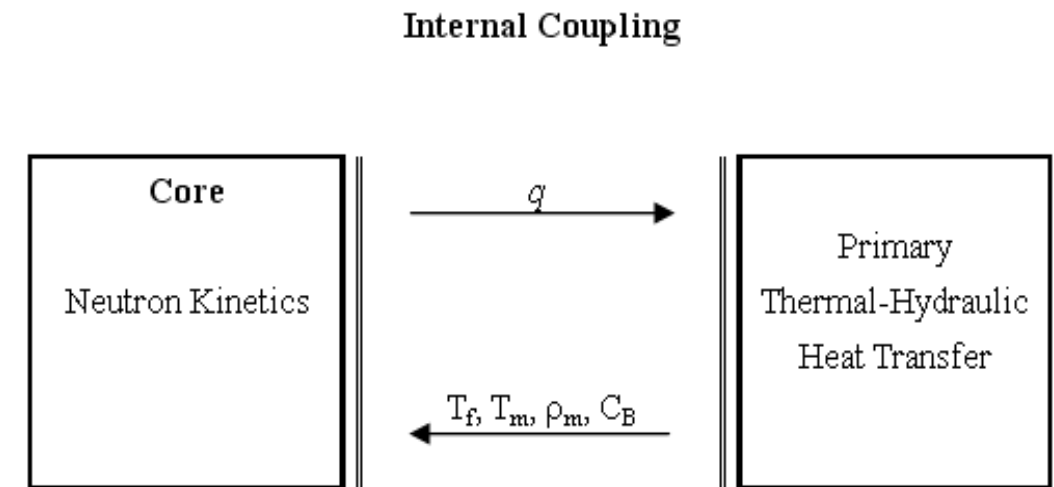
Development – involves choice and implementation of three basic components:

- *Way of coupling & Coupling approach*
- *Spatial coupling & Temporal coupling*
- *Coupling numerics & Coupled convergence*

WAY OF COUPLING

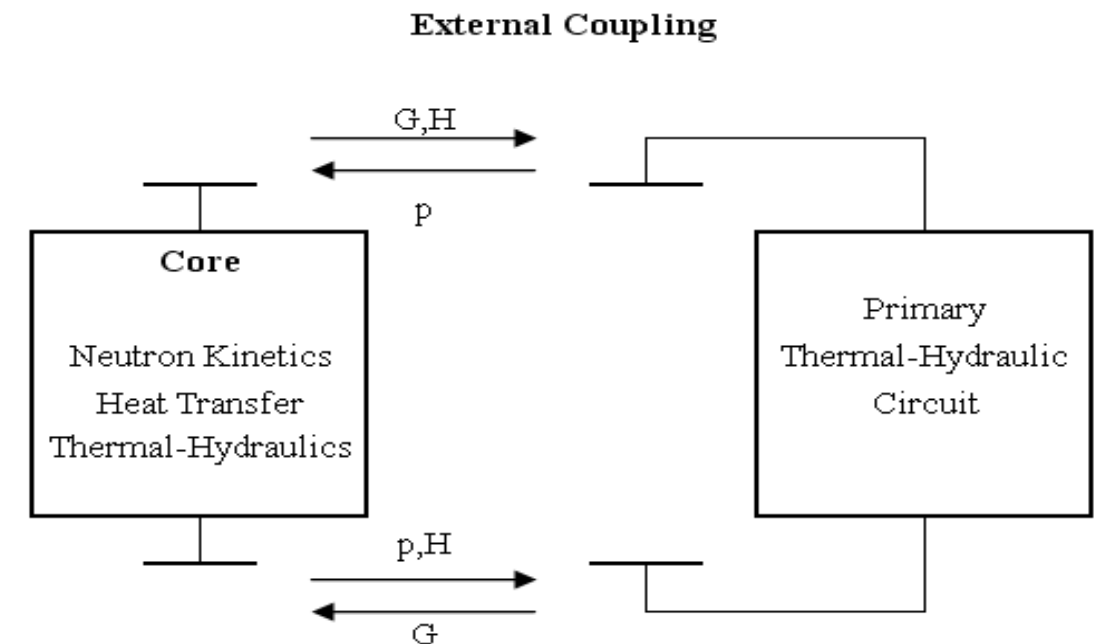
Internal coupling – closely coupled 3-D neutron kinetics with core heat-transfer and conduction model of a thermal-hydraulic code:

!! *Much more consistent and sophisticated but involves significant code modifications.*



External coupling – coupled core and the balance of plant within the thermal-hydraulics modeling:

!! *Few exchanged parameters and little code modification but unstable and slow to converge.*



COUPLING APPROACH

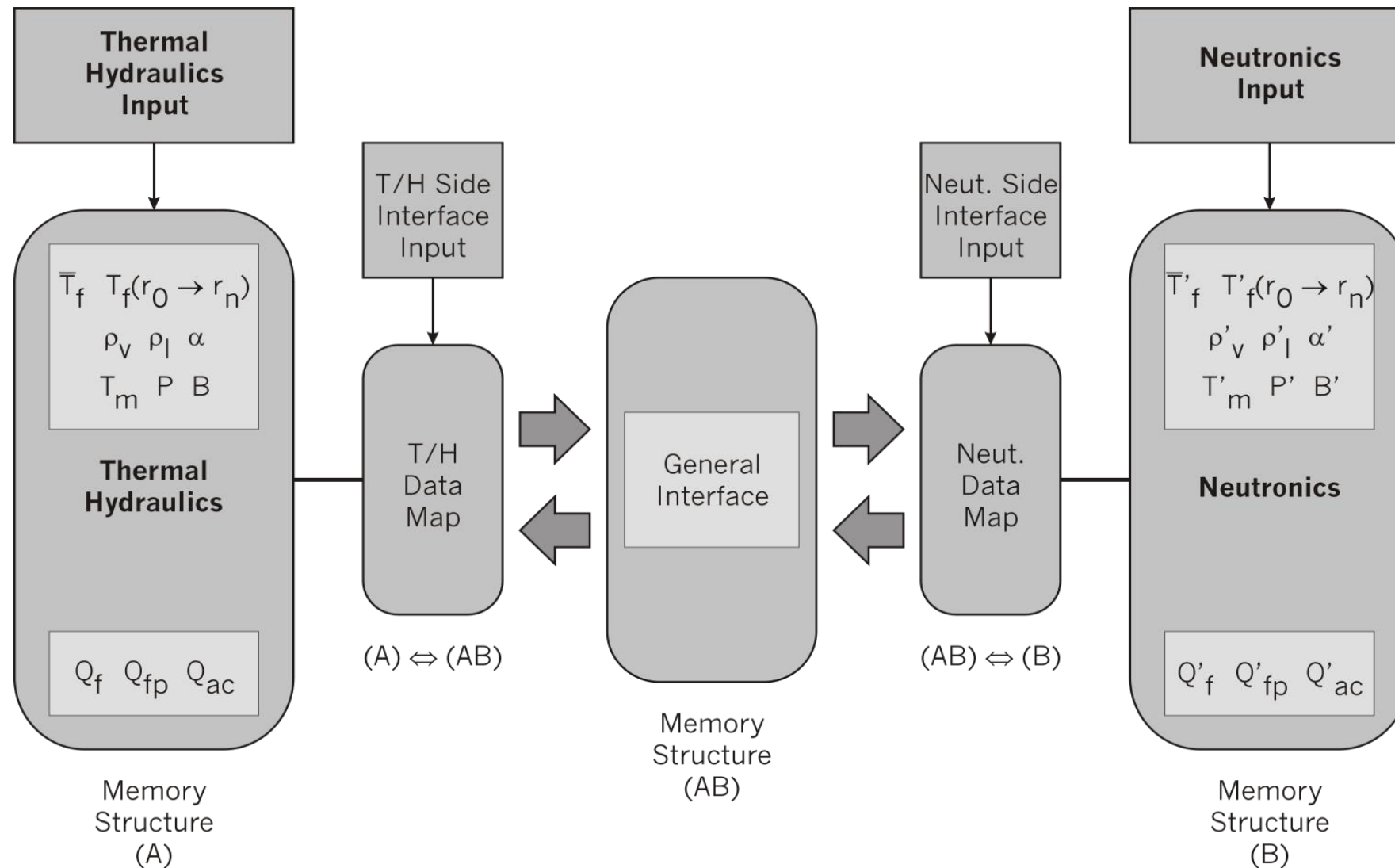
Integration algorithm

- The neutronics code is implemented as a routine into the thermal-hydraulic code usually using the same restart capabilities.
- Extensive code changes for complete code integration are required.
- Sequential execution of thermal-hydraulic and neutronics modules is performed.

Parallel processing schemes (PVM or MPI)

- Minimum code changes.
- Allows the codes to be run separately and exchanges data during the calculation.
- Provides opportunity to improve coupled code efficiency.
- The thermal-hydraulic and neutronics modules have separate restart capabilities.

COUPLING APPROACH

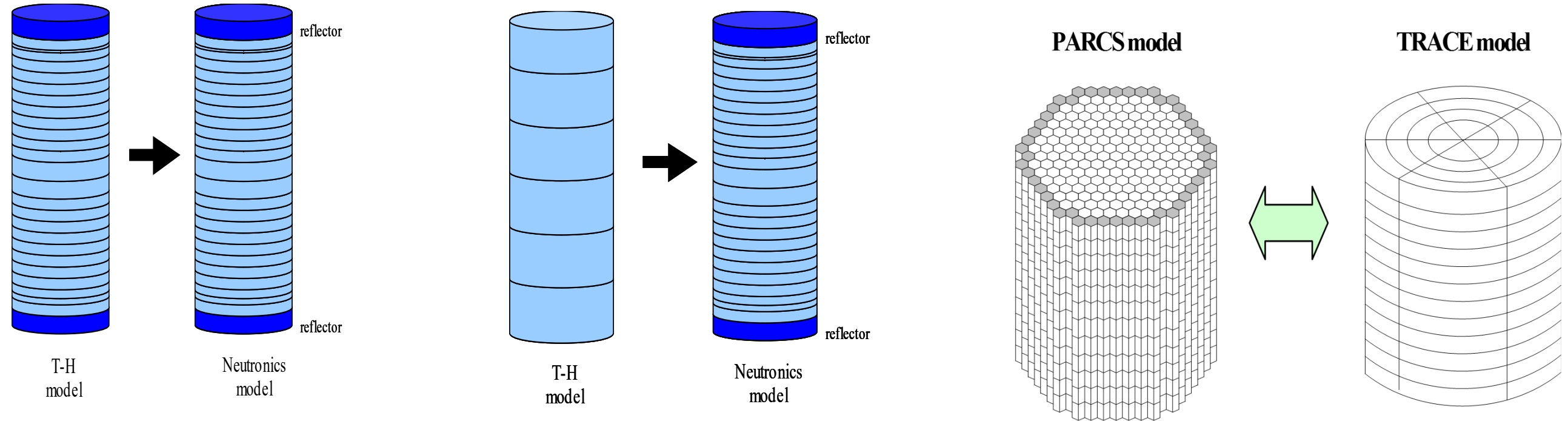


- 1998
PARCS / GI / RELAP
Parallel Virtual Machine
- 1999
PARCS / GI / TRAC-M
- 2000
Merging GI into PARCS
(Memory Copy instead of PVM)
Automatic Mapping for PARCS / TRAC-M
- 2004
Merging PARCS into TRACE
Static Linking Library
No PVM

SPATIAL COUPLING

- The spatial coupling means spatial mesh overlays (mapping schemes) between:
 - Thermal-hydraulics core model;
 - Heat structure/rod model;
 - Neutronics model.
- Each of these models has its own spatial nodalization (mesh layout).
- The purpose of mapping schemes is to arrive to mesh overlays between the nodalizations of the different models.
- Fixed spatial coupling: usually one thermal-hydraulic channel represents one neutronics assembly.
- Flexible spatial coupling: user's specified mapping schemes:
 - Accuracy vs. efficiency and capabilities.
 - Depends on the nature of the transient to be analyzed.
 - Depends also on the type of core thermal-hydraulic model.
 - Needs to be done in both axial direction and radial plane.

SPATIAL COUPLING



- Choosing proper spatial mesh overlays for given transient to provide accurate solutions in a reasonable time is a challenging task:
 - The exact detailed mapping provides better spatial resolution in coupled calculations;
 - This refinement requires significant computation resources;
 - The computer requirements become enormous for transients with coupled core/system interactions.

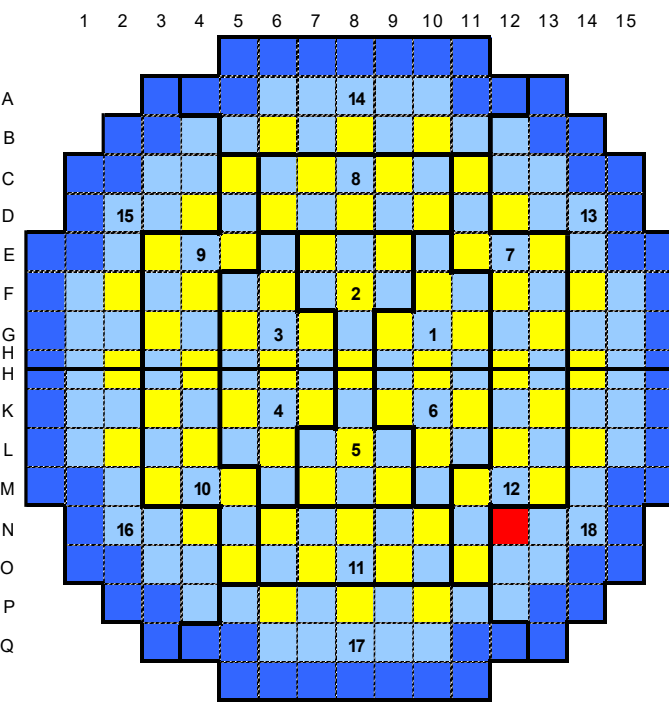
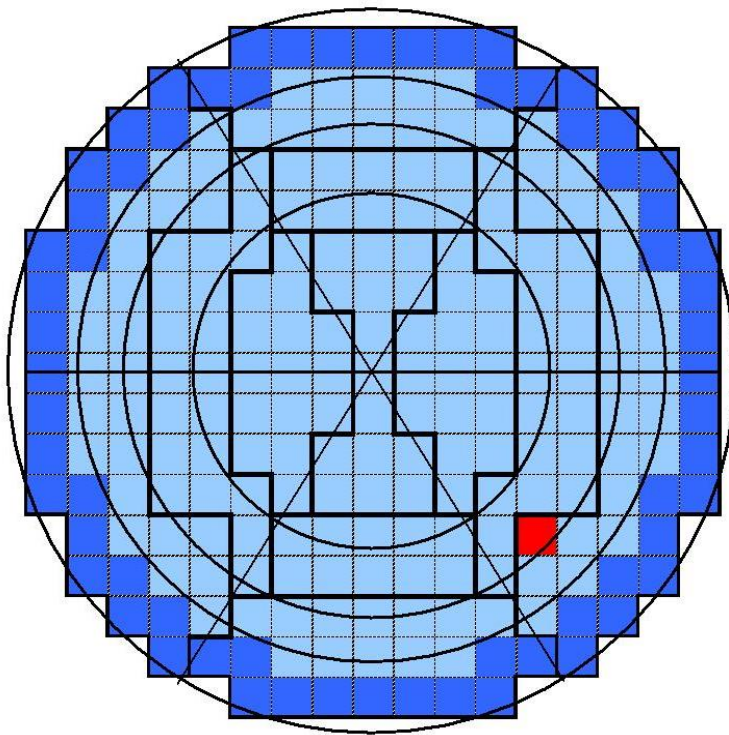
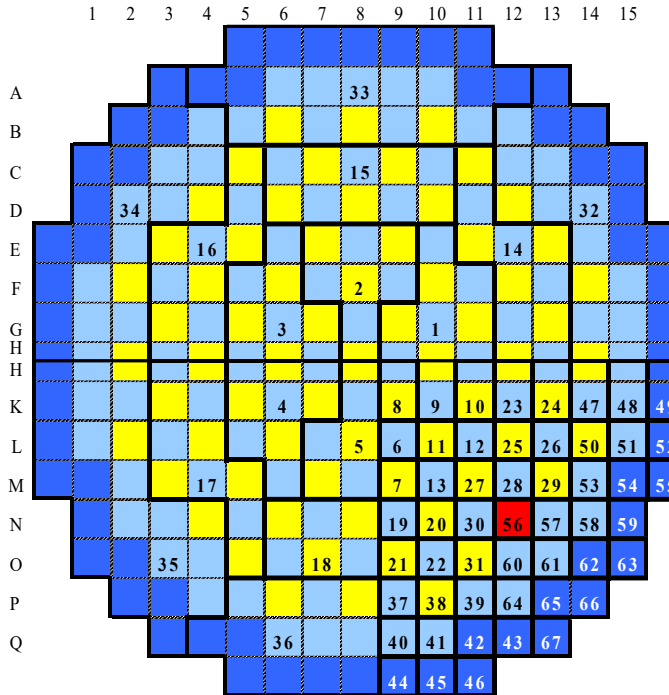
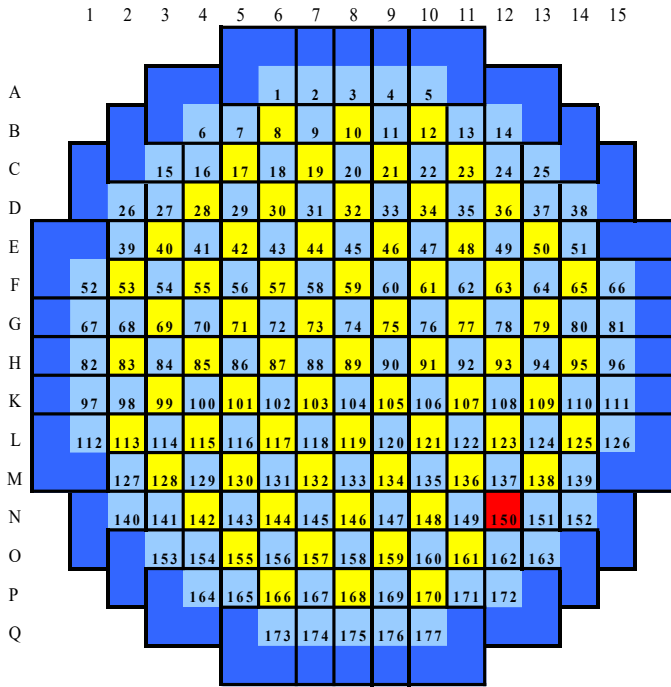
!!! More important for local parameters.

SPATIAL COUPLING - PWR SPATIAL MESH OVERLAYS

PWR Rod Ejection Accident (REA) studies

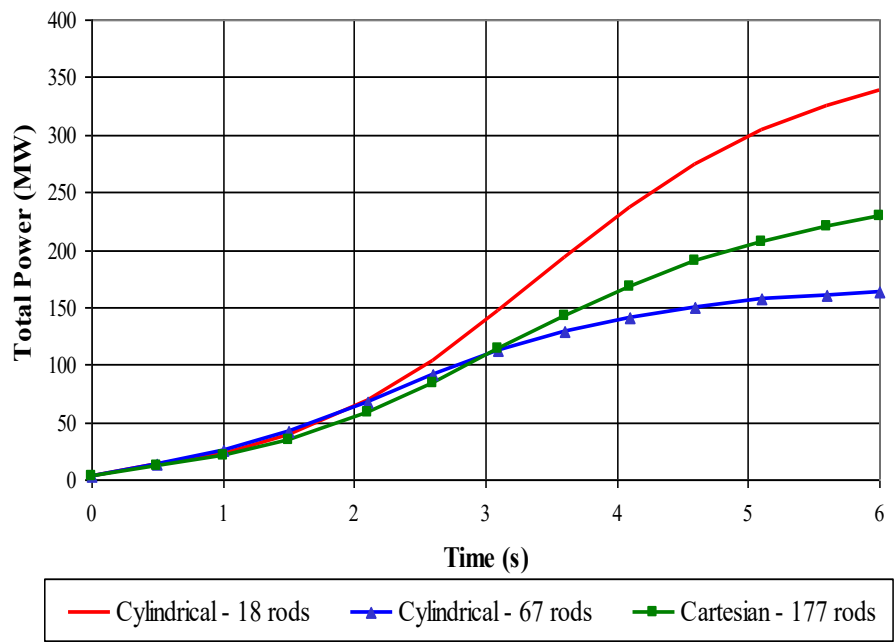
- The REA simulations are sensitive to the radial spatial coupling schemes.
- The impact of neutronics mesh refinement is well known.
- The local and global predictions are sensitive to the thermal-hydraulic refinement (only the local refinement of the Doppler (fuel temperature) feedback model does not necessarily improve accuracy of the results).
- The obtained results indicate that the thermal-hydraulic feedback is non-linear.

SPATIAL COUPLING - PWR SPATIAL MESH OVERLAYS

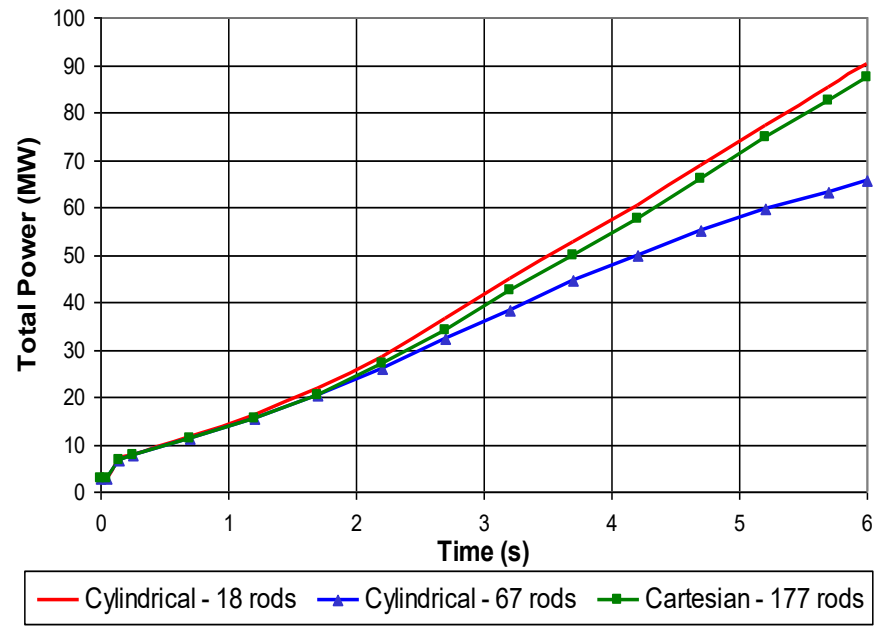


SPATIAL COUPLING - PWR SPATIAL MESH OVERLAYS

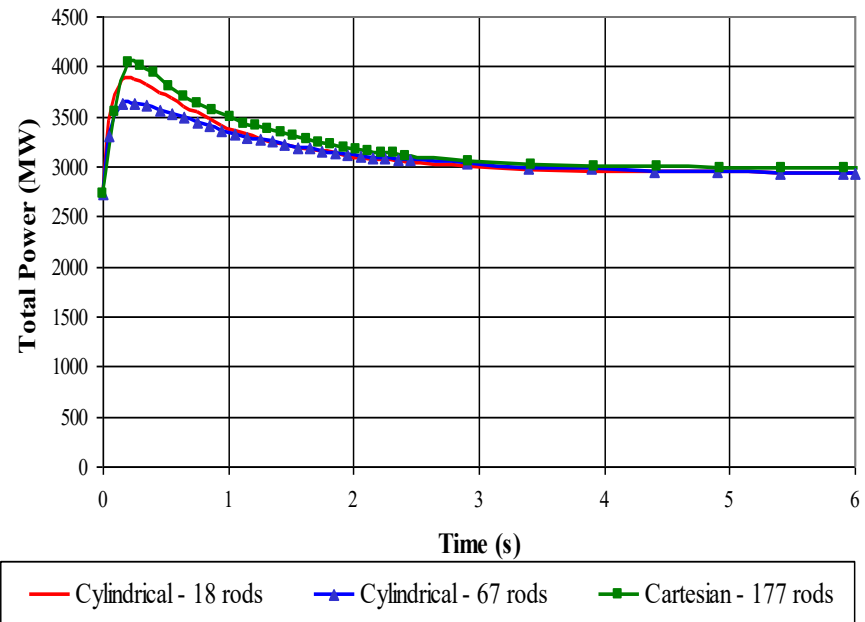
BOC HZP



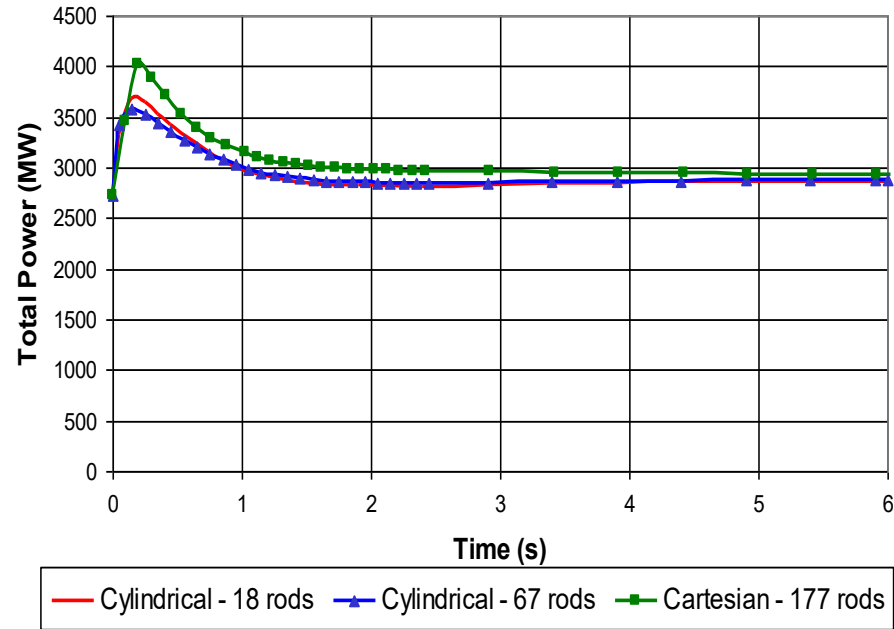
EOC HZP



BOC HFP



EOC HFP



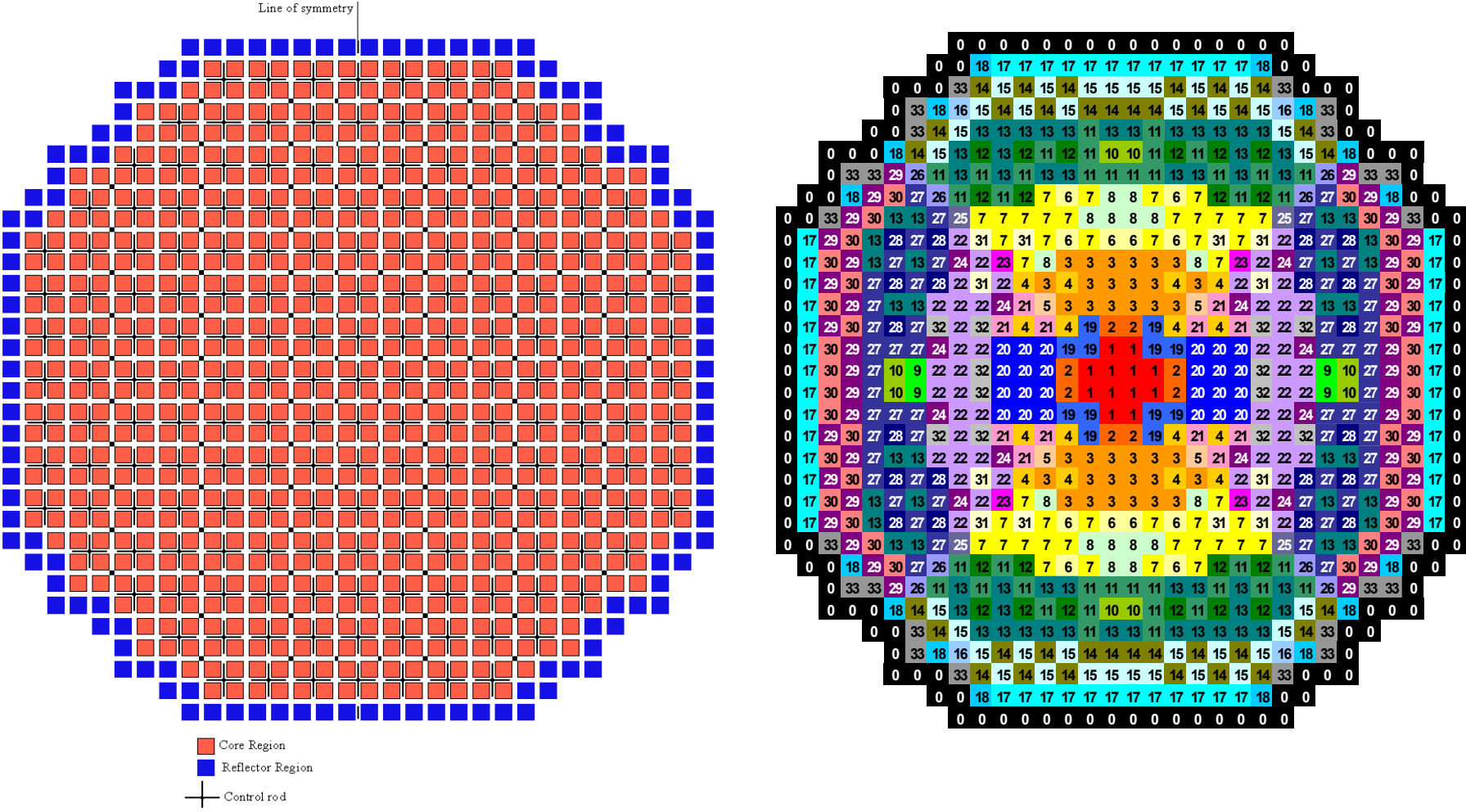
SPATIAL COUPLING - BWR SPATIAL MESH OVERLAYS

BWR spatial mesh overlays

- BWR core contains a large number of fuel assemblies (~800): *the exact detail coupled thermal-hydraulic and neutronics modeling of such core requires significant computational resources.*
- Thus, there is a great interest in optimization of coupled calculations: *computational costs could be reduced when collapsing similar assemblies into a single thermal-hydraulic channel while keeping the detail neutronics modeling.*
- When mapping neutronics assemblies to thermal-hydraulic channels usually different rules are followed:
 - For example, usually similar neutronics assemblies in terms of their design are mapped in one channel;
 - Other characteristics used to lump together several neutronics assemblies to a thermal-hydraulic channel are the relative power, coolant flow, void distribution, burnup, and etc.

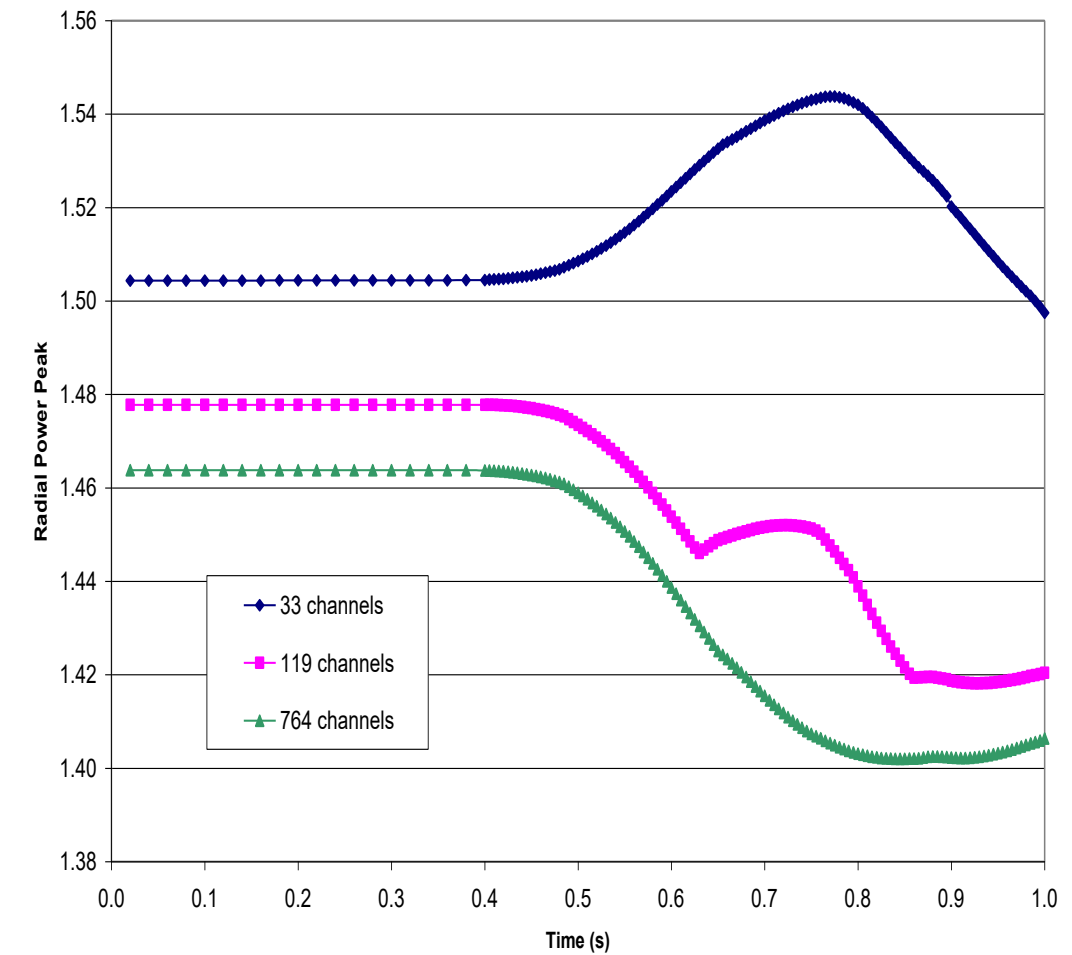
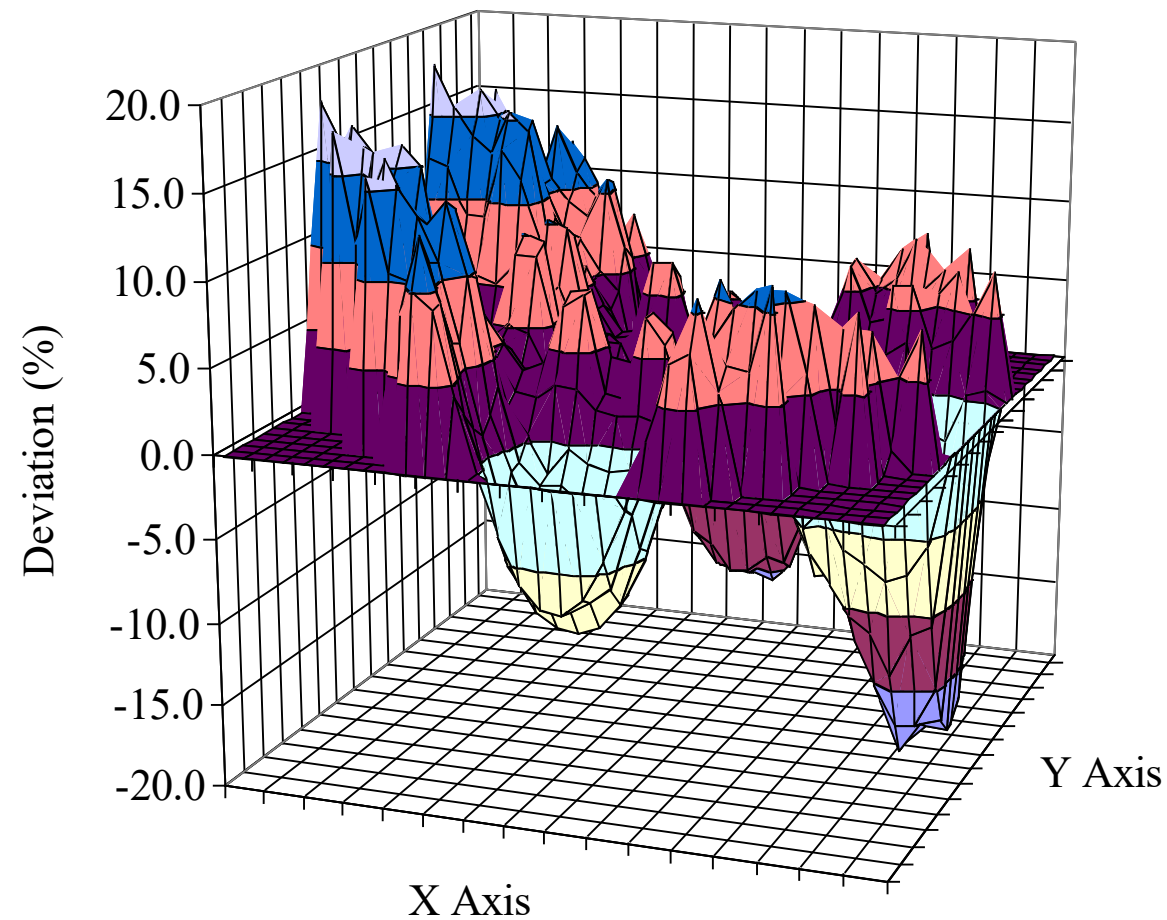
SPATIAL COUPLING - BWR SPATIAL MESH OVERLAYS

Peach Bottom 2 core model and 33-channel scheme used for BWR
Turbine Trip Benchmark



SPATIAL COUPLING - BWR SPATIAL MESH OVERLAYS

Deviations between different mapping schemes for initial conditions and during the transient



TEMPORAL COUPLING

- The temporal coupling and synchronization is very important since the neutron kinetics and thermal-hydraulics codes usually have their own time step selection algorithms:
3-D kinetics variable time step algorithm + meshing scheme with the thermal-hydraulic time step selection algorithm.
- For any successive time integration step, it is required that the maximum change be smaller than a user-specified tolerance ε :
 ε_G – global, set up to limit global power change to 1% per time step
 ε_L – local, set up to limit local flux change to 10% per time step
- The global and local changes are calculated by evaluating the first derivative in time of the monitored variables (power and flux).

TEMPORAL COUPLING - EXAMPLE

- The required kinetics time step size is determined as follows:

$$T_G^K = \frac{\varepsilon_G}{d\varepsilon_G/dt} \quad ; \quad T_L^K = \frac{\varepsilon_L}{d\varepsilon_L/dt} \quad ; \quad T_{estimate}^K = \min(T_G^K, T_L^K) \geq T_{min}^K ,$$

where T_{min}^K is equal to the thermal-hydraulic time step size.

- The final time step size is:

$$T_{final}^K = \min(T_{estimate}^K, T_{max}^K) ,$$

where T_{max}^K is specified by the user.

COUPLING NUMERICS

- The numerical schemes of multi-physics coupling can be classified by three types: **(1) Explicit;** **(2) Semi-implicit;** and **(3) Fully implicit.**
- The classification is based on timing of exchange of the neutronics and thermal-hydraulics information.

The thermal-hydraulics model:

- Computes new moderator and fuel properties and sends feedback parameters to the neutronics model – T_m , ρ_m , α , C_b and T_f .
- Uses neutronics power as heat source for conduction.

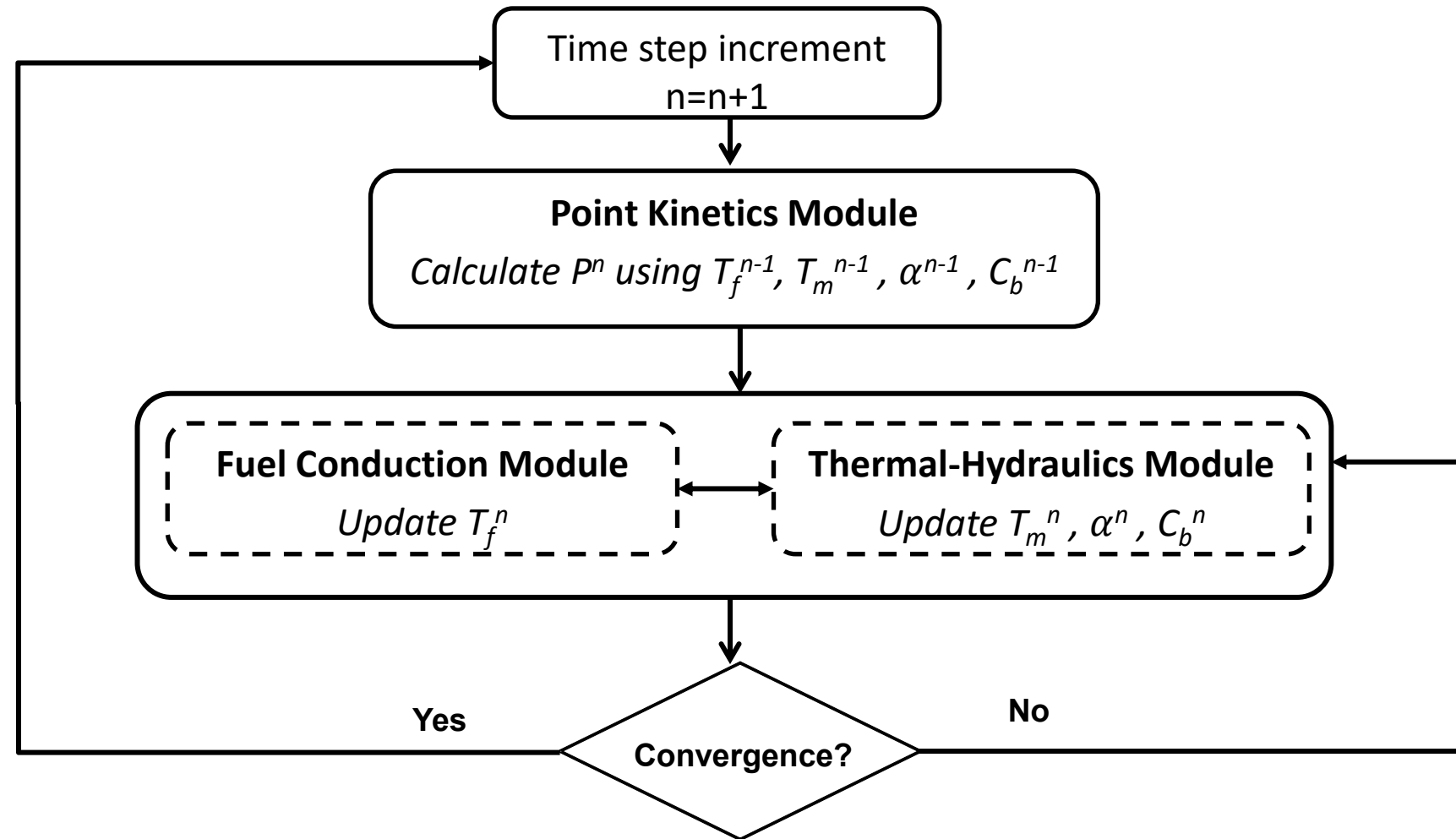
The neutronics model:

- Uses the feedback parameters to update cross-sections based on local node conditions.
 - Computes 3-D power distributions and sends node-wise power distribution to the TH model.
- The major drawback of the traditional numerical coupling schemes is that they sequentially solve coupled fields using “marching” methods.
 - This restricts the time step size or take considerable CPU time if have to converge for each time step.
 - In novel developments the coupled fields are solved simultaneously; thereby increasing the time step size.

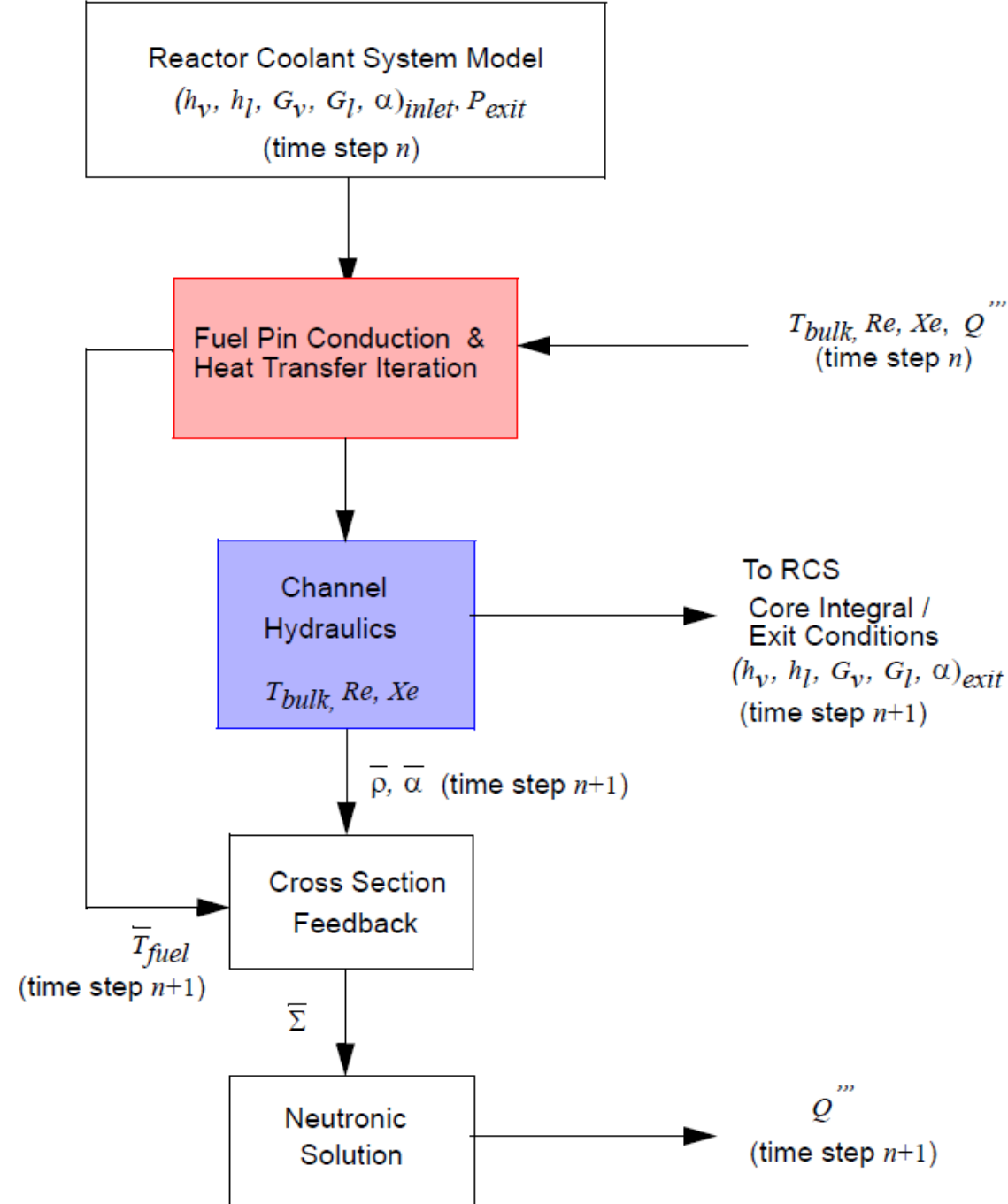
CROSS-SECTION MODELING

- Modeling cross-section variations for off-nominal core conditions is an important issue in the coupled 3-D neutronics/thermal-hydraulic transient simulations:
 - The cross-section modeling for coupled 3-D simulations is based on the data generated in the so-called base and branch calculations using a lattice physics code and assembled in a cross-section library.
 - The cross-section libraries include macroscopic cross-sections, diffusion coefficients, ADFs, form functions and CDFs, microscopic Xe cross-sections, and kinetics parameters (inverse neutron velocities and delayed neutron parameters).
 - For transient applications, it is important the modeling of cross-term dependencies in cross-section behavior.

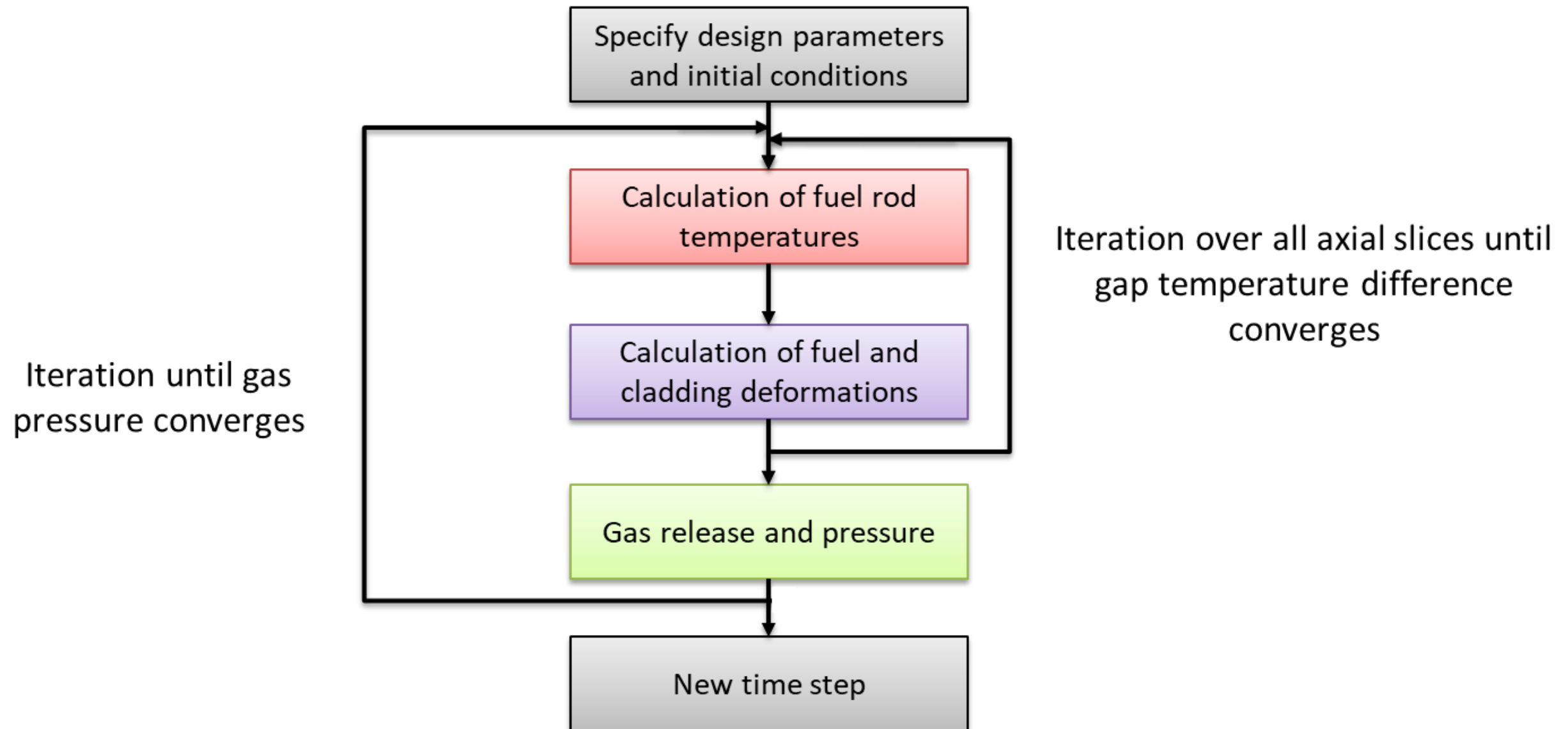
POINT KINETICS MODEL IN TH CODES (EXAMPLE)



1-D TH MODELS IN NEUTRONICS CORE SIMULATORS (EXAMPLE)



1.5D COUPLING BETWEEN THERMAL, MECHANICAL AND PHYSICO-CHEMICAL MODELLING OF FUEL (EXAMPLE)



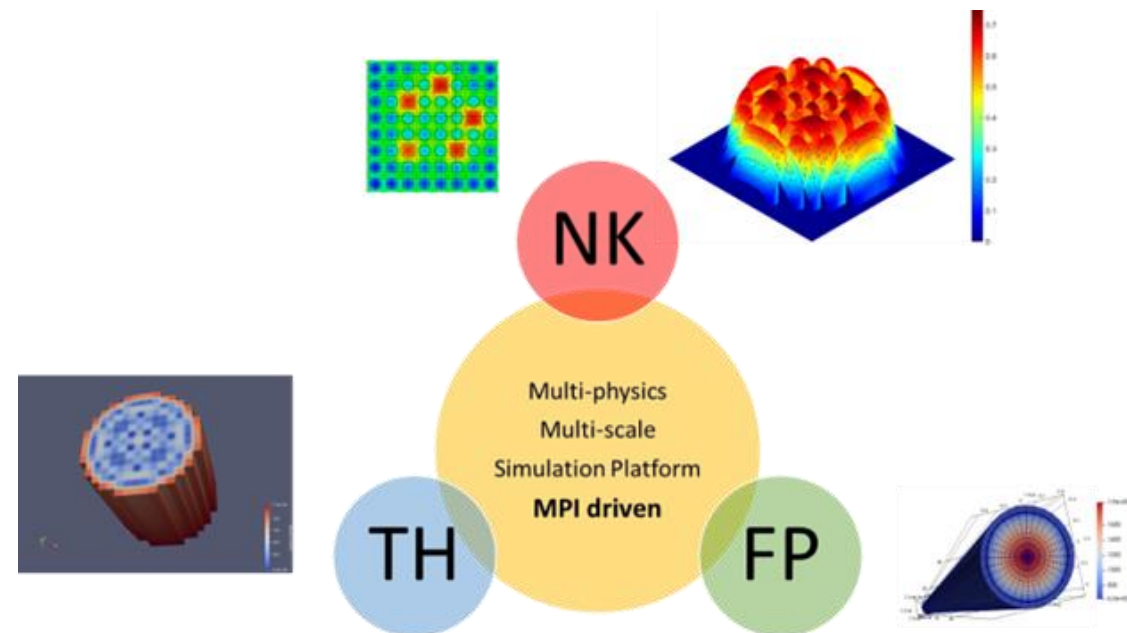
TRADITIONAL LWR ANALYSIS AND SIMULATION - SUMMARY

- Efficient (based on nodal core physics methodologies) fuel management with an acceptable accuracy:
 - !! *Detailed 3-D core neutronics models*
 - &
 - !! *Simplified core thermal-hydraulics models.*
- Limited by available computation power: earlier safety methods utilized a simple kinetics response:
 - !! *System codes*
 - &
 - !! *in conjunction with a conservative approach to bind fuel operation.*

TRADITIONAL LWR ANALYSIS AND SIMULATION - SUMMARY

- Sequential system/core calculations.
- System thermal-hydraulics codes provide BC to local core response models:
 - !! *Sub-channel codes.*
- Limited to transient calculations where the feedback effects are small.
- Incorporating full 3-D models of the reactor core into system transient codes allows for a “best-estimate” calculation of interactions in time and space between both:
 - !! *Core neutronics and thermal-hydraulics models*
 - &
 - !! *Core behaviour and plant dynamics.*

MPI SERVER-CLIENT COUPLING INTERFACE FOR EXTERNAL COUPLING

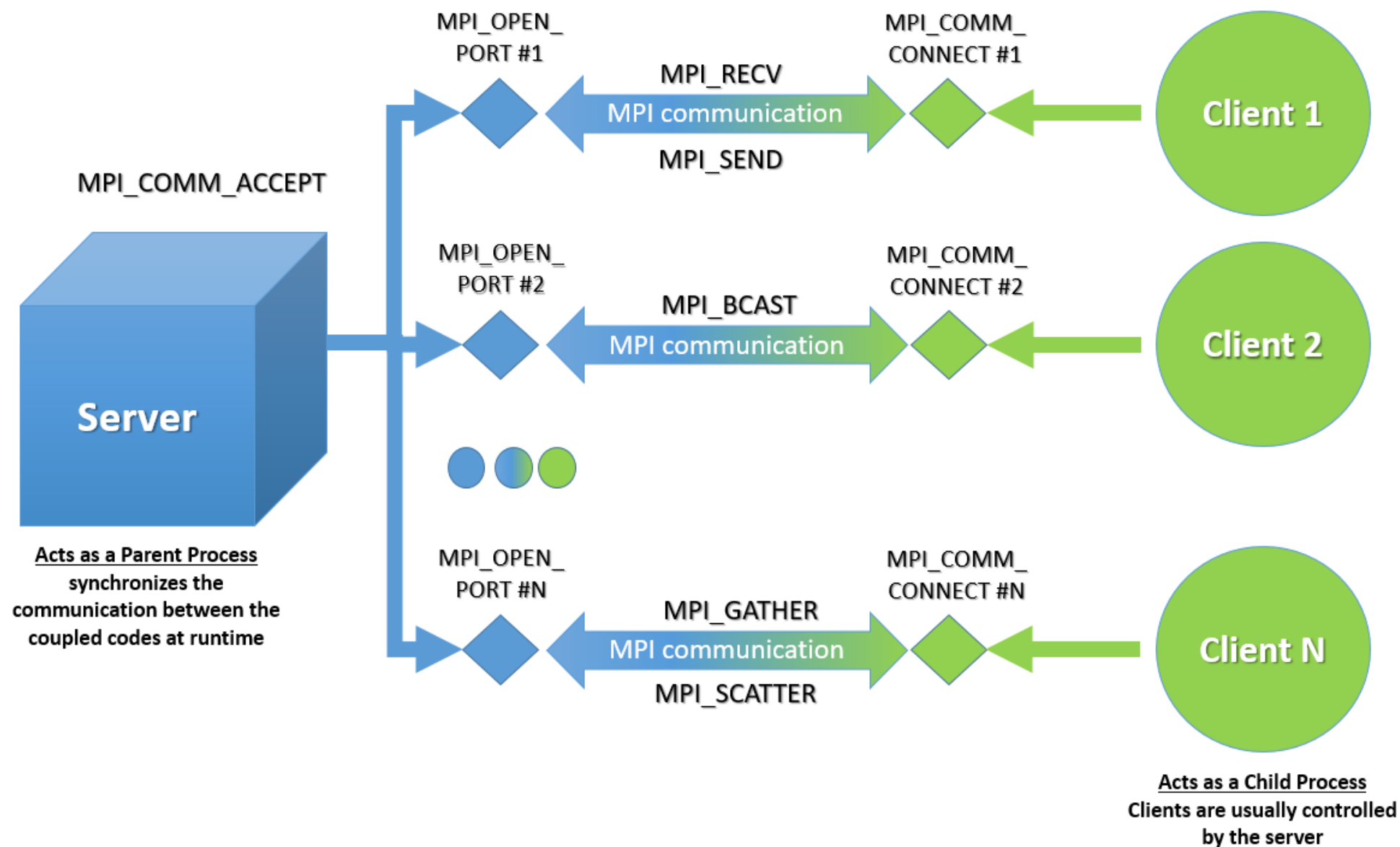


MPI Coupling Interface

- One of the desired features of a multi-physics coupling interface is to allow for flexible coupling algorithm that unfold the full capacities of each code.
- Coupling Interfaces (CI) based on a MPI Server-Client (S-C) communication algorithm (loose coupling) are often selected for simulation platforms.
- The different components of the platform incorporate a S-C interface to unlock a bidirectional communication between them.
- Examples: MPICH, OpenMPI, etc.
- The MPI based external coupling scheme permits the communication between the different simulation codes without extended modifications in their compilation projects (just adding the coupling interface source file).

MPI Coupling Interface

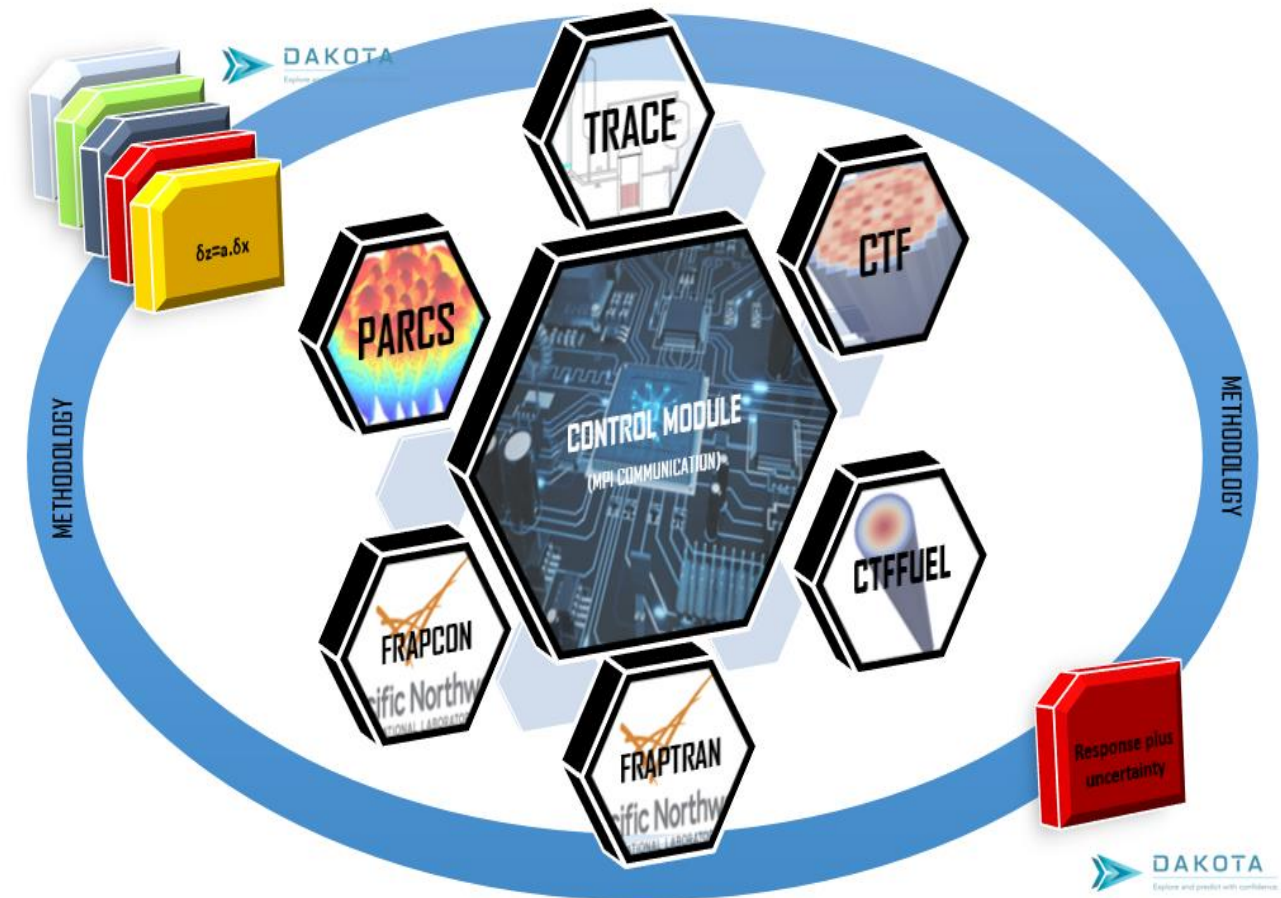
Control module and MPI Server/Client coupling interface



Multi-physics multi-scale coupling interface

TRACE/CTF/PARCS coupling

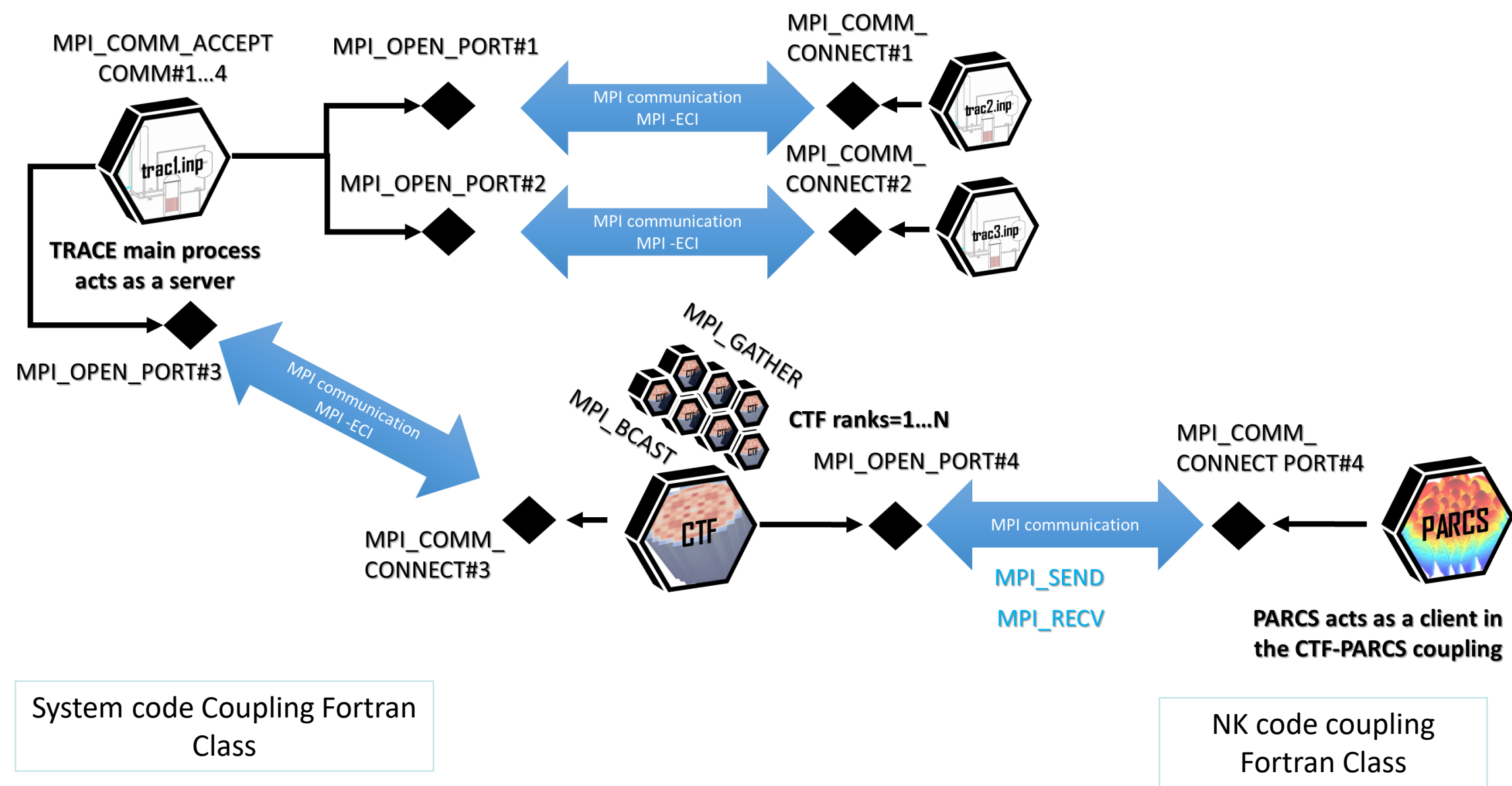
- Based on US NRC codes, for **Best Estimate Plus Uncertainty (BEPU)** safety analysis of LWRs.
- MPI coupling interface.
- Coupling between CTF, TRACE and PARCS.
- Hi2Lo model information from FRAPCON/FRAPTRAN (FAST) to CTF/CTFFuel.
- Methodology for the BEPU analysis.



Multi-physics multi-scale coupling interface

TRACE/CTF/PARCS coupling

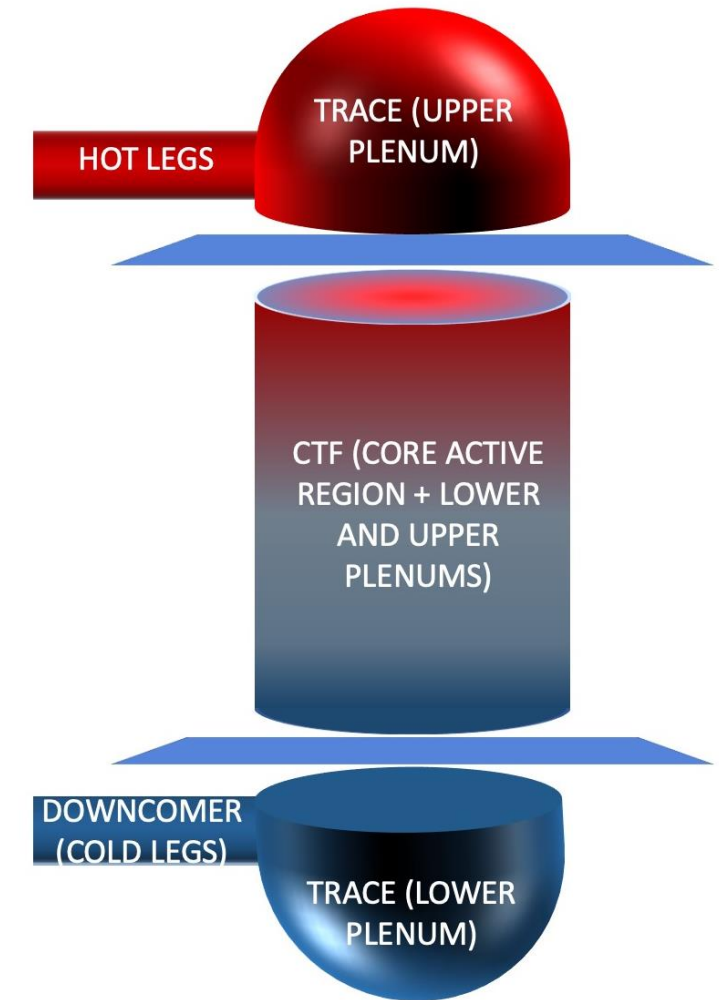
- MPI coupling algorithm based on the Server-Client communication.



Multi-physics multi-scale simulation platform

TRACE / CTF Coupling Domain Decomposition

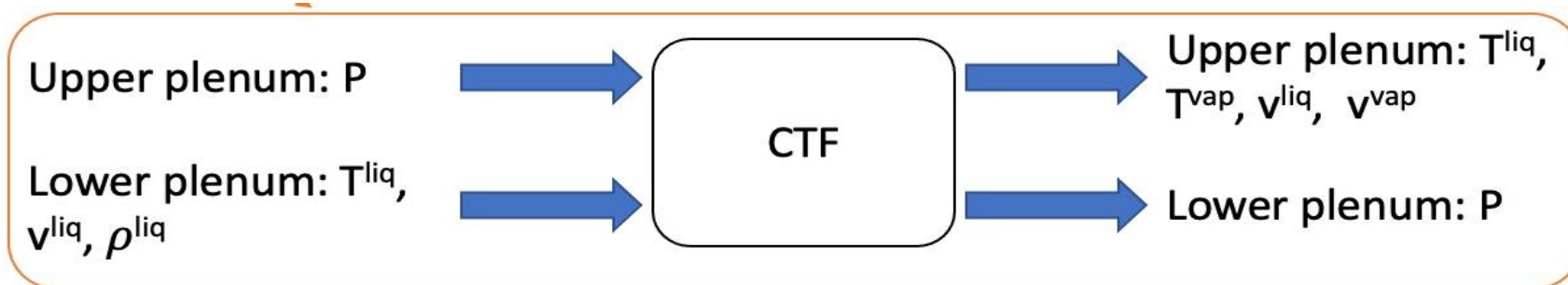
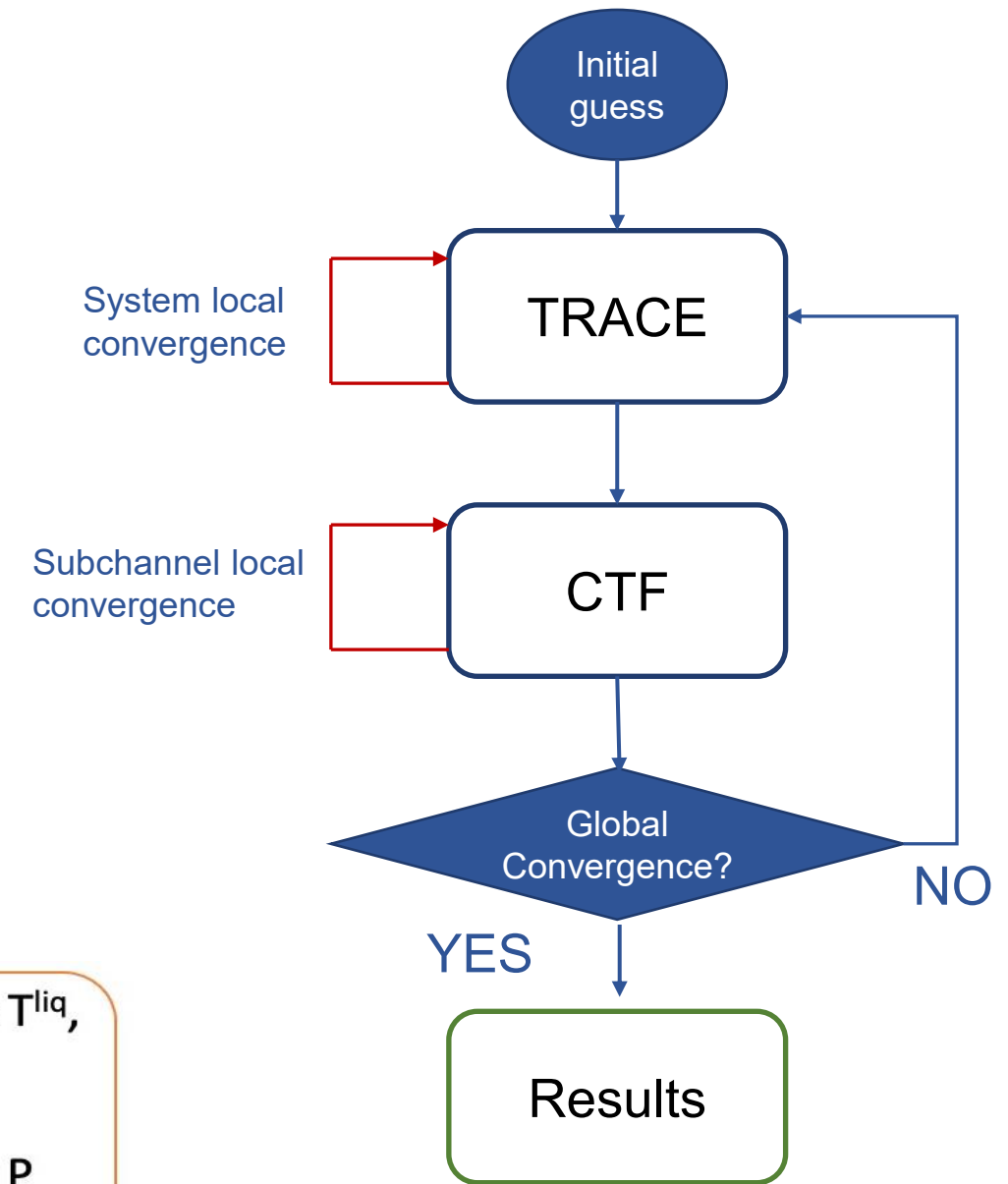
- External coupling based on the MPI version of the TRACE ECI.
- Picard Iteration coupled convergence scheme in steady-state.
- Explicit and semi-implicit schemes used in temporal coupling for transients.
- CTF models the reactor core active region and small regions of the upper lower and upper plenums.
- TRACE models the three coolant loops, the complementary of lower and upper plenums, and the downcomer.
- The coupling is done by exchanging boundary conditions at the bottom and the top of the CTF model.
- The smallest time-steps size between both codes is being used.



Multi-physics multi-scale simulation platform

TRACE / CTF Coupling: Steady-state simulation scheme

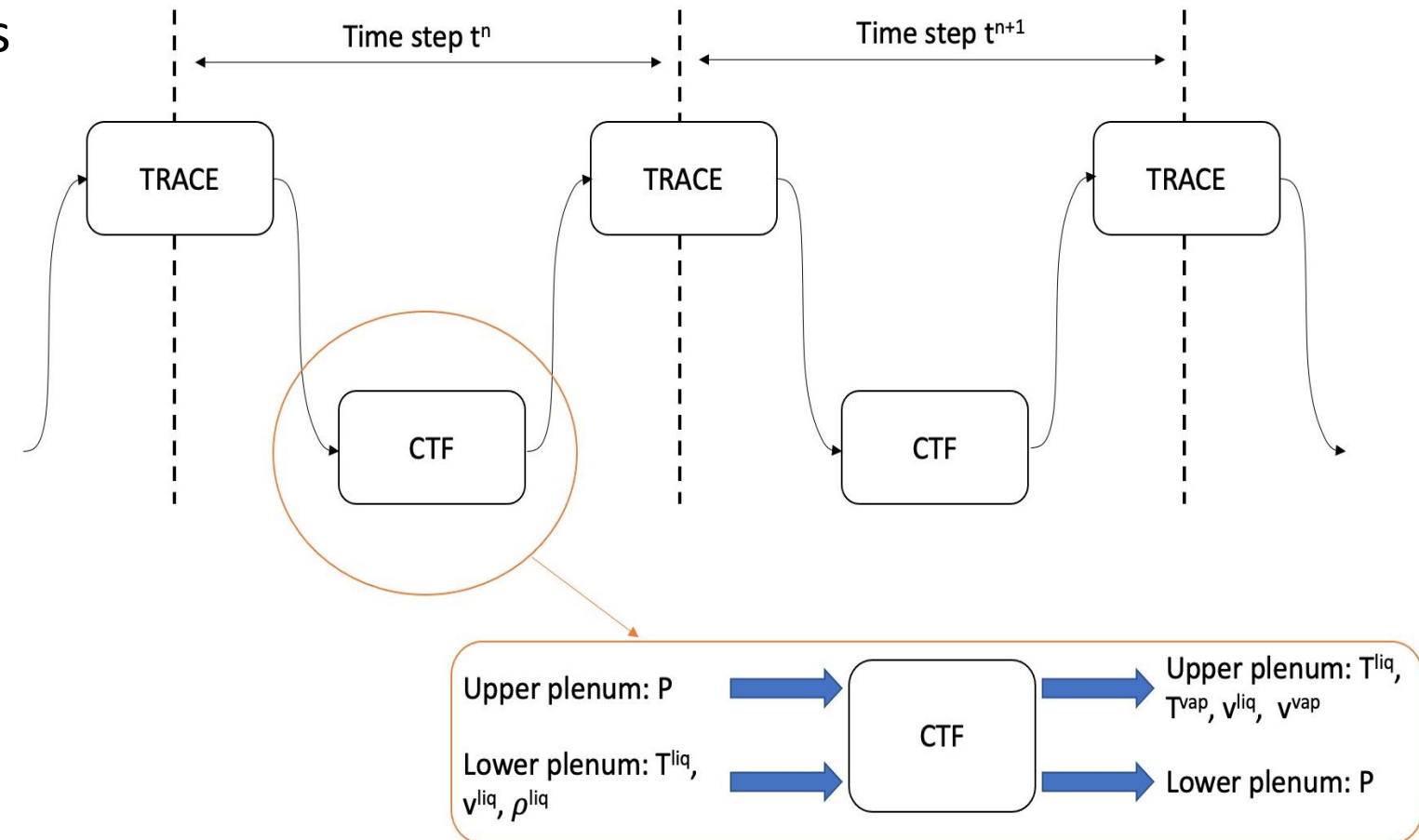
- Picard iteration scheme:
 - TRACE begins the simulation running a steady-state (coupled step t^n).
 - Feedback information send from TRACE to CTF and the domain interfaces.
 - CTF runs a steady state with updated core boundary conditions (coupled step t^n).
 - Check global convergence criteria (feedback parameters residuals).
 - Information send back from CTF to TRACE.
 - TRACE runs a steady-state (coupled step t^{n+1}).



Multi-physics multi-scale simulation platform

TRACE / CTF Coupling: transient simulation scheme

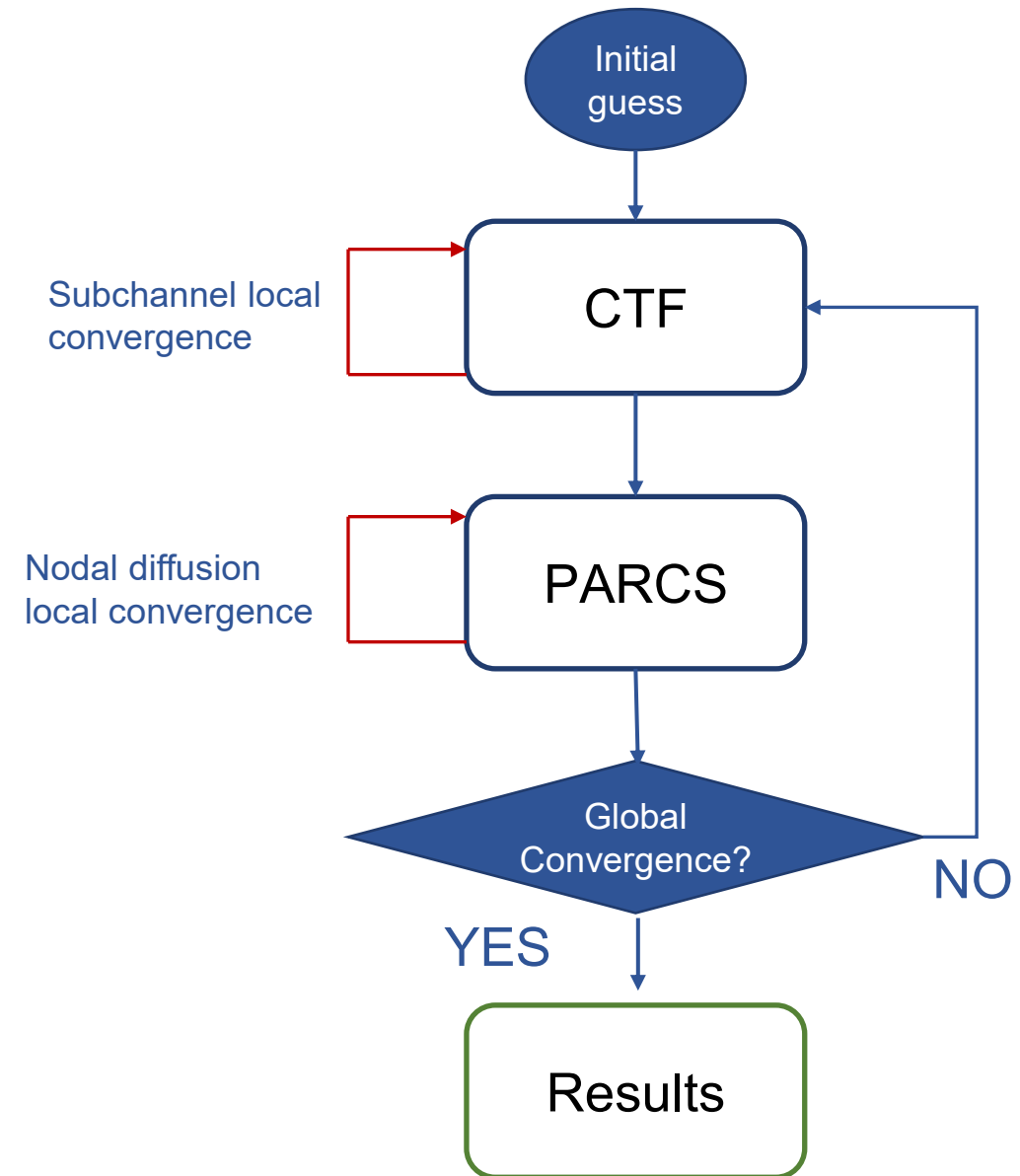
- Explicit scheme for temporal coupling.
- The transfer of information between codes is being done using MPI communication protocol.
- Iteration scheme for transients:
 - TRACE beginning of time step t^n
 - Information send from TRACE to CTF
 - CTF runs time step t^n with new boundary conditions
 - Information send from CTF to TRACE
 - TRACE runs time step t^{n+1} with new boundary conditions.



Multi-physics multi-scale simulation platform

CTF / PARCS Coupling: steady simulation scheme

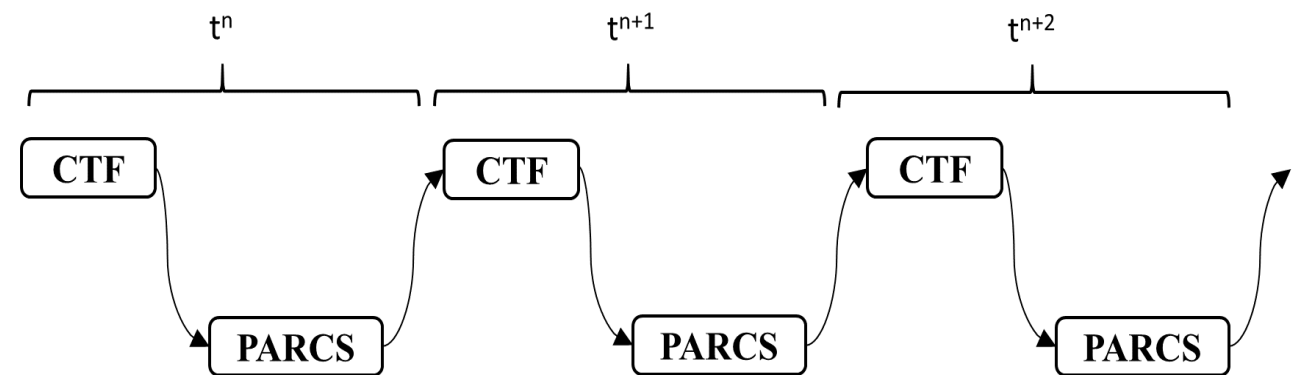
- Picard iteration scheme:
- The transfer of information between codes is being done using MPI communication protocol between CTF and TRACE/PARCS.
- The iteration scheme is:
 - CTF begins the simulation running a steady-state (coupled step t_n)
 - Feedback information send from CTF to PARCS
 - CTF runs a steady state with updated linear power in the fuel rods (coupled step t_n)
 - Check global convergence criteria (feedback parameters residuals, l-2 and l-inf)
 - TH information send back from CTF to PARCS
 - PARCS runs a steady-state (coupled step t_{n+1})
 - ...
 - Save linear powers at the end of the steady-state.



Multi-physics multi-scale simulation platform

CTF/PARCS Coupling: transient simulation scheme

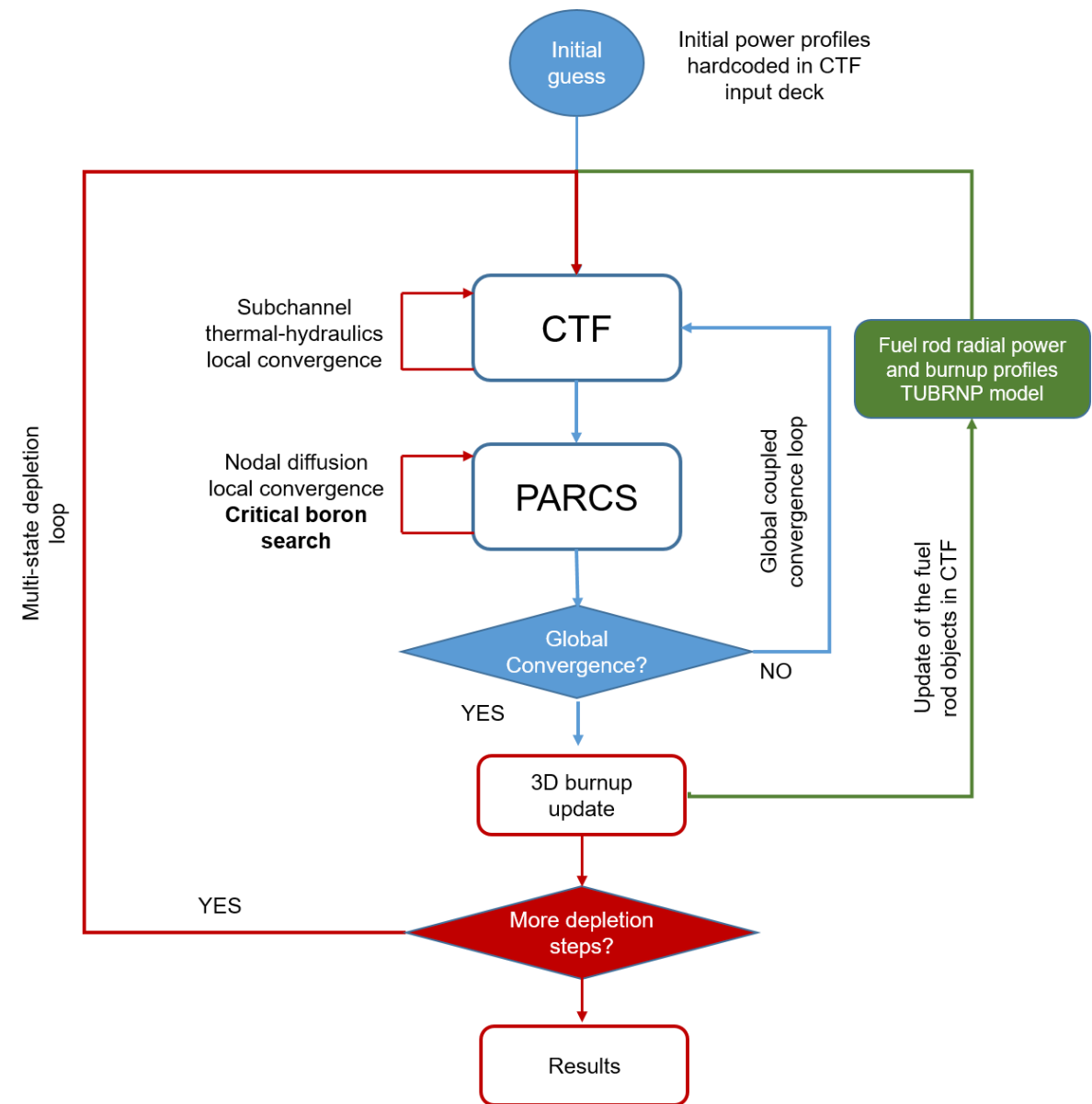
- Explicit scheme for temporal coupling.
- The transfer of information between codes is being done using MPI communication protocol between CTF and TRACE/PARCS.
- Iteration scheme for transients:
 - CTF beginning of time step t^n
 - Feedback information send from CTF to PARCS
 - PARCS runs time step t^n with new TH update
 - Power profiles and total power send back from PARCS to CTF
 - CTF runs time step t^{n+1} with new boundary conditions.



Multi-physics multi-scale simulation platform

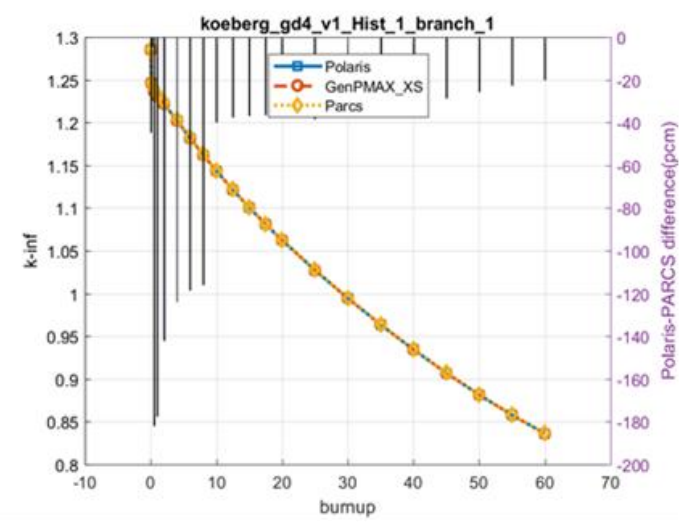
CTF / PARCS Coupling: depletion mode

- Equivalent to the steady-state coupling scheme but for a multi-state simulation.
- PARCS calculates critical boron concentration for every defined depletion step. Then, CTF is not providing those values.
- Updating by fuel rod burnup in CTF after each depletion step.
- Computing new pellet power and burnup profiles with the depletion module after each depletion step.

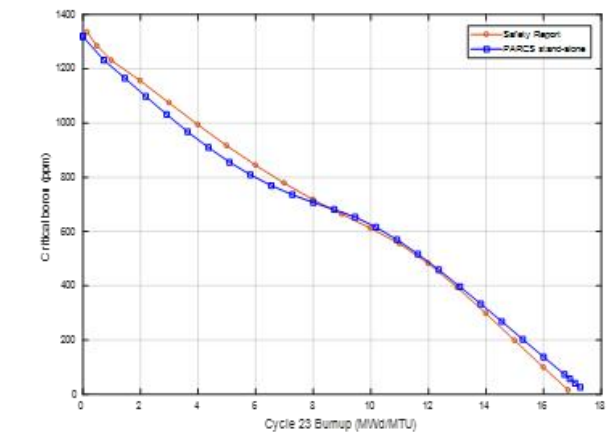


Multi-physics multi-scale simulation platform

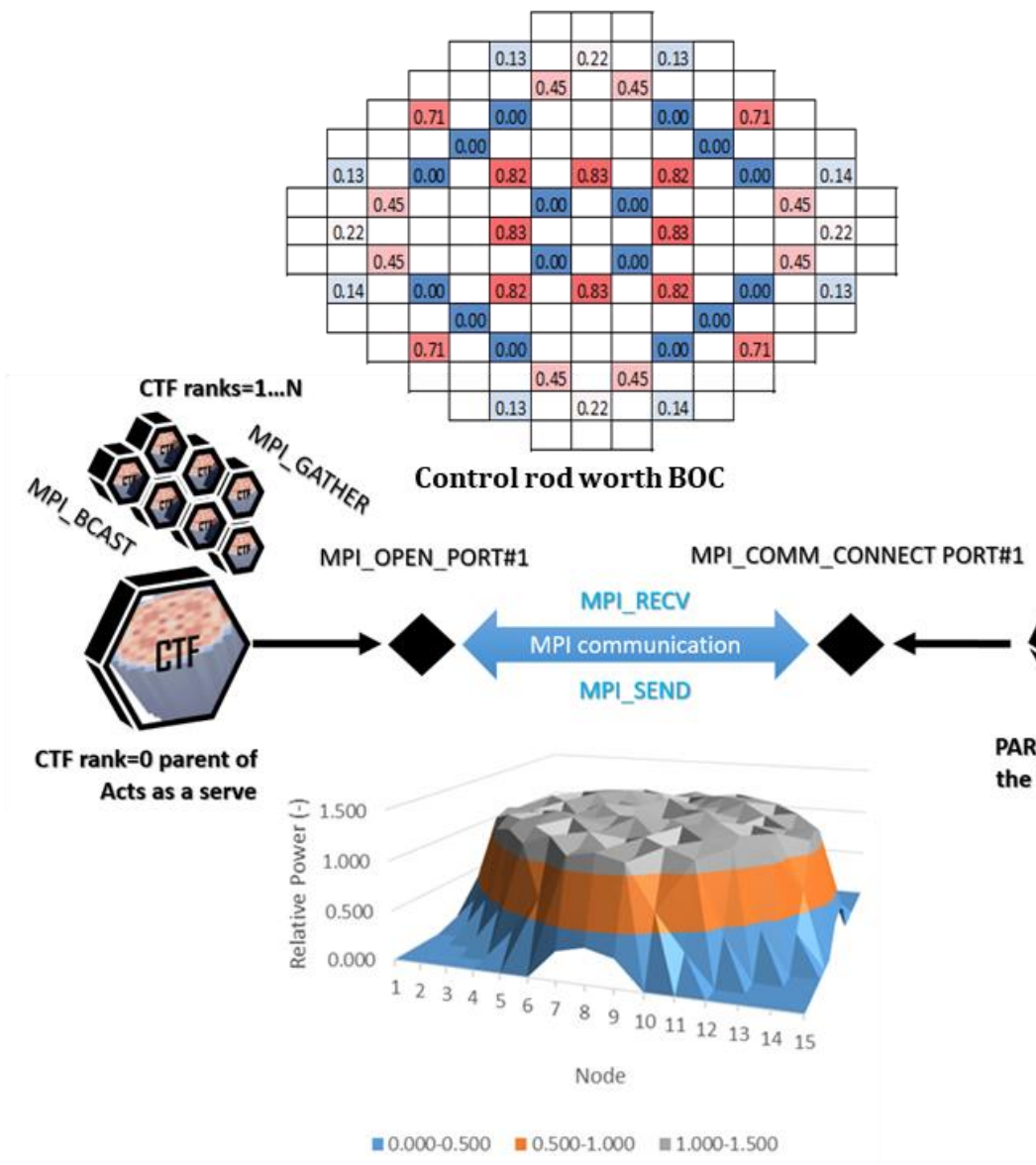
CTF/PARCS Coupling: demonstration results



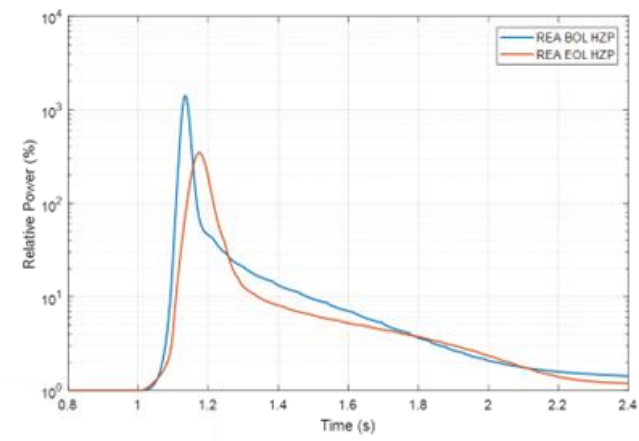
Cross section verification



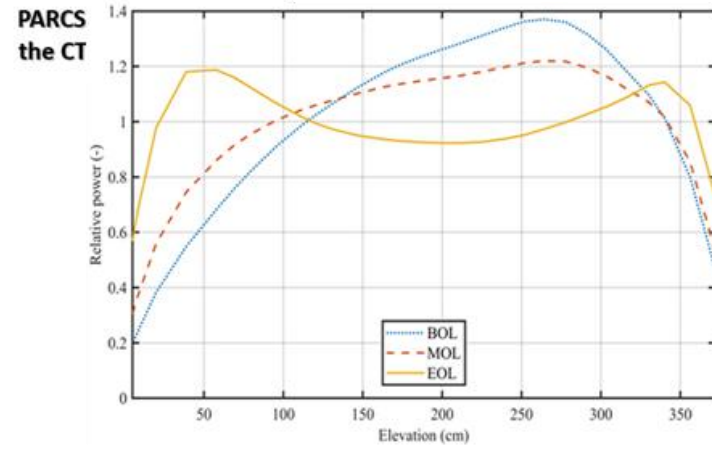
Critical boron vs burnup cycle 23



Core radial power shape BOC



Control rod ejection BOC vs EOC



Axial power profiles during Cycle 23

Next Class: *Point Kinetics and 3D Nodal Kinetics Models in Thermal-Hydraulic Codes*